

Home Office Network Setup Documentation

Group 9

Each of us played a vital role in ensuring the successful implementation of the home office network setup.

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-

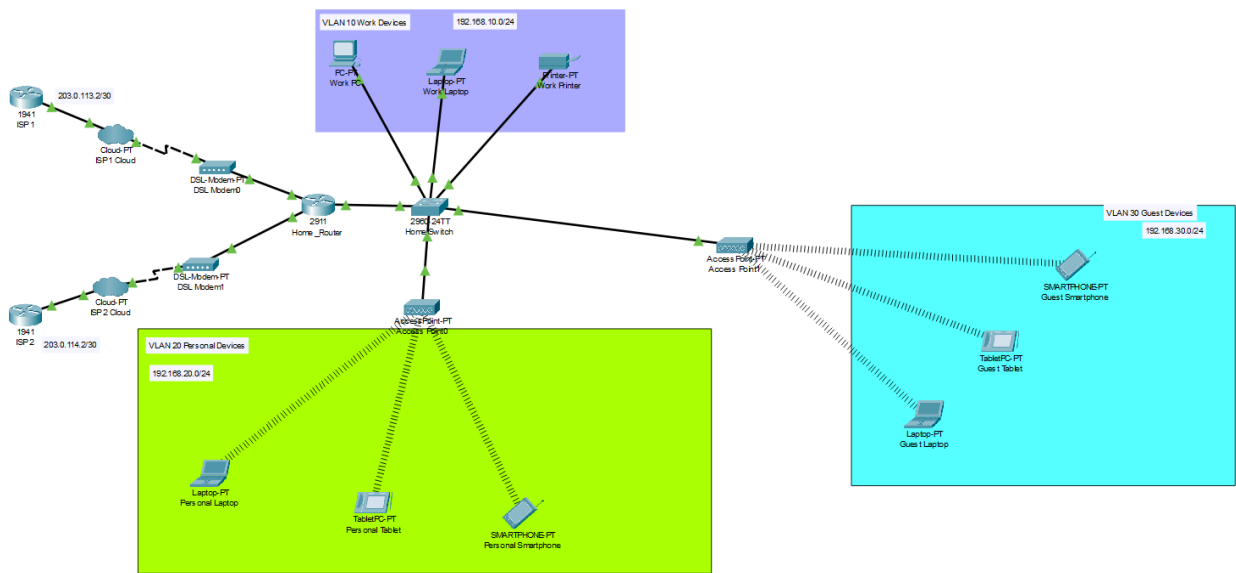
This documentation provides a comprehensive overview of the network setup designed for a home office environment, including both high-level and low-level details. These documents serve as a complete guide to both the design and implementation of the home office network, ensuring that it meets the client's requirements for performance and security.

High-Level Documentation for Home Office Network Configuration

Overview

This document provides a guide to setting up a home office network using Cisco Packet Tracer. The setup includes VLAN configuration, inter-VLAN routing, and basic security measures.

Cisco Packet Tracer Logical View



Bill of Materials (BOM)

Device Type	Device Name	IP Address	VLAN	Purpose
Router	Home Router	WAN G0/0 203.0.113.1	N/A	This router provides inter-VLAN routing, acting as the gateway for all VLANs

		LAN G0/1 192.168.0.0		(Work, Personal, Guest).
Router	ISP	G0/0 203.0.113.2/30	N/A	Simulates the Internet Service Provider connection.
DSL Modem	Home DSLModem	N/A	N/A	Connects Home Router to ISP
WAN Emulation	Cloud0	N/A	N/A	Simulates ISP Connectivity
Switch	Home Switch	N/A	VLAN 10, 20, 30	This switch manages the VLANs and port allocations for connected devices.
PC	PC	DHCP	VLAN 10	Connected to VLAN 10 (Work Devices) and receives network settings via DHCP.
Laptop	Work Laptop	DHCP	VLAN 10	Connected to VLAN 10 (Work Devices) and receives network settings via DHCP.
Printer	Printer	DHCP	VLAN 10	Connected to VLAN 10 (Work Devices) and receives network settings via DHCP.
Laptop	Personal Laptop	DHCP	VLAN 20	Connected to VLAN 20 (Personal Devices), receives network settings via DHCP.
Tablet	Personal Tablet	DHCP	VLAN 20	Connected to VLAN 20 (Personal Devices), receives network settings via DHCP.
Smartphone	Personal Smartphone	DHCP	VLAN 20	Connected to VLAN 20 (Personal Devices), receives network settings via DHCP.
Laptop	Guest Laptop	DHCP	VLAN 30	Connected to VLAN 30 (Guest Devices), receives network settings via DHCP.
Tablet	Guest Tablet	DHCP	VLAN 30	Connected to VLAN 30 (Guest Devices),

				receives network settings via DHCP.
Smart Phone	Guest SmartPhone	DHCP	VLAN 30	Connected to VLAN 30 (Guest Devices), receives network settings via DHCP.
Access Point	Access Point (Personal)	N/A	VLAN 20	Provides wireless connectivity to personal devices (e.g., laptops, tablets, smartphones) on VLAN 20.
Access Point	Access Point (Guest)	N/A	VLAN 30	Provides wireless connectivity to guest devices (e.g., laptops, tablets, smartphones) on VLAN 30.

I. Switch configurations

1. Initial Setup

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Home_Switch
Home_Switch(config)# line console 0
Home_Switch(config-line)# password Group9
Home_Switch(config-line)# login
Home_Switch(config-line)# end
```

Setting the hostname and securing console access.

2. Enable Secret and VTY Lines

```
Home_Switch#configure terminal
Home_Switch(config)#enable secret OpenDay
Home_Switch(config)#line vty 0 15
Home_Switch(config-line)#password CyberGirl
```

```
Home_Switch(config-line)#login
Home_Switch(config-line)#end
```

Configuring enable secret and securing VTY lines for remote access.

3. Encrypt Passwords and Set MOTD Banner

```
Home_Switch#configure terminal.
Home_Switch(config)#service password-encryption
Home_Switch(config)#banner motd #AUTHORIZED ACCESS ONLY#
```

Encrypting passwords and setting a Message of the Day (MOTD) banner.

4. VLAN Configuration

```
Home_Switch(config)#vlan 10
Home_Switch(config-vlan)#name Work_Devices_VLAN
Home_Switch(config-vlan)#vlan 20
Home_Switch(config-vlan)#name Personal_Devices_VLAN
Home_Switch(config-vlan)#vlan 30
Home_Switch(config-vlan)#name Guest_VLAN
Home_Switch(config-vlan)#exit
```

Creating VLANs for different device group

5. Assign Ports to VLANs

```
Home_Switch(config)#interface range FastEthernet0/1-10
Home_Switch(config-if-range)#description Connection to VLAN 10 Work Devices
Home_Switch(config-if-range)#switchport mode access
Home_Switch(config-if-range)#switchport access vlan 10
Home_Switch(config-if-range)#exit
```

```
Home_Switch(config)#interface range FastEthernet0/11-20
Home_Switch(config-if-range)#description Connection to VLAN 20 Personal
Devices
Home_Switch(config-if-range)#switchport mode access
Home_Switch(config-if-range)#switchport access vlan 20
Home_Switch(config-if-range)#interface range FastEthernet0/21-24
Home_Switch(config-if-range)#description Connection to VLAN 30 Guest
Devices
Home_Switch(config-if-range)#switchport mode access
Home_Switch(config-if-range)#switchport access vlan 30
Home_Switch(config-if-range)#exit
```

Assigning switch ports to the respective VLANs.

6. Configure Trunk Ports

```
Home_Switch(config)#interface GigabitEthernet0/1
Home_Switch(config-if)#description connection to Home_Router
Home_Switch(config-if)#switchport mode trunk
Home_Switch(config-if)#switchport trunk allowed vlan 10,20,30
Home_Switch(config)#interface GigabitEthernet0/2
Home_Switch(config-if)#description connection to connection to access point
Home_Switch(config-if)#exit
```

Configuring trunk ports to allow VLAN traffic between the router and the switch.

7. Save Configuration

```
Home_Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Saving the switch configuration.

II. Router Configuration

1. Initial Setup

```
Router>enable
Router#configure terminal
Router(config)#hostname Home_Router
Home_Router(config)#enable secret OpenDay
Home_Router(config)#line console 0
Home_Router(config-line)#password Group9
Home_Router(config-line)#login
Home_Router(config-line)#exit
Home_Router(config)#line vty 0 4
Home_Router(config-line)#password CyberGirl
Home_Router(config-line)#login
Home_Router(config-line)#transport input ssh telnet
Home_Router(config-line)#exit
Home_Router(config)#banner motd #UNAUTHORIZED ACCESS IS PROHIBITED#
```

Setting the hostname, securing access, and configuring a MOTD banner.

2. Interface Configuration

```
Home_Router#configure terminal
Home_Router(config)#interface GigabitEthernet0/0
Home_Router(config-if)#description connection to ISP_1
Home_Router(config-if)#ip address 203.0.113.1 255.255.255.252
Home_Router(config-if)#no shutdown
Home_Router(config-if)#exit
Home_Router(config)#interface GigabitEthernet0/2
Home_Router(config-if)#description connection to ISP_2
Home_Router(config-if)#ip address 203.0.114.1 255.255.255.252
```

```
Home_Router(config-if)#no shutdown
Home_Router(config-if)#exit

Home_Router(config)#interface GigabitEthernet0/1
Home_Router(config-if)#description connection to Home_Switch
Home_Router(config-if)#no shutdown
Home_Router(config-if)#exit
```

Configuring physical interfaces on the router.

3. Sub-Interface Configuration for VLANs

```
Home_Router(config)#interface GigabitEthernet0/1.10
Home_Router(config-subif)#encapsulation dot1Q 10
Home_Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Home_Router(config-subif)#exit

Home_Router(config)#interface GigabitEthernet0/1.20
Home_Router(config-subif)#encapsulation dot1Q 20
Home_Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Home_Router(config-subif)#exit

Home_Router(config)#interface GigabitEthernet0/1.30
Home_Router(config-subif)#encapsulation dot1Q 30
Home_Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Home_Router(config-subif)#exit
```

Configuring sub-interfaces for VLAN routing.

4. DHCP Configuration for VLANs

1. Create DHCP Pools

In this configuration, separate DHCP pools are created for the Work Devices VLAN, Personal Devices VLAN, and Guest VLAN, allowing each group of devices to receive appropriate IP settings based on their network segment.

- **Work Devices VLAN:**

Configures a DHCP pool for the Work Devices VLAN (VLAN 10) on the router. It assigns IP addresses from the **192.168.10.0/24** network and sets the default gateway and DNS server for devices in this VLAN.

```
Router(config)# ip dhcp pool Work_Devices_VLAN
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

- **Personal Devices VLAN:**

Configures a DHCP pool for the Personal Devices VLAN (VLAN 20). This pool uses the **192.168.20.0/24** network and assigns the gateway and DNS server to personal devices.

```
Router(config)# ip dhcp pool Personal_Devices_VLAN
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

- **Guest VLAN:**

Sets up a DHCP pool for the Guest VLAN (VLAN 30), using the **192.168.30.0/24** network. It specifies the default gateway and DNS server for guest devices.

```
Router(config)# ip dhcp pool Guest_VLAN
Router(dhcp-config)# network 192.168.30.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.30.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

2. Save Router Configuration:

```
Home_Router#copy running-config startup-config
```

5. Set Up Wireless Encryption (WPA2-PSK) for Secure Wireless Connections:

Configures wireless security for the Personal Devices VLAN (SSID: Personal_Devices_VLAN) and Guest VLAN (SSID: Guest_VLAN) using WPA2-PSK encryption.

- **VLAN 20 SSID:** Personal_Devices_VLAN

WPA2-PSK Passphrase: 0p3nD@Y@uGu5t#15

- **VLAN 30 SSID:** Guest_VLAN

WPA2-PSK Passphrase: Op3nD@Y15#GrP!91\$

6. Access Control Lists (ACLs) Configuration

- Deny traffic from 192.168.10.0/24 to 192.168.20.0/24. Deny traffic from 192.168.10.0/24 to 192.168.30.0/24. Permit all other traffic from 192.168.10.0/24 to any destination.
- Deny traffic from 192.168.20.0/24 to 192.168.30.0/24. Permit all other traffic from 192.168.20.0/24 to any destination
- Deny traffic from 192.168.30.0/24 to 192.168.10.0/24. Deny traffic from 192.168.30.0/24 to 192.168.20.0/24. Permit all other traffic from 192.168.30.0/24 to any destination

- D. The permit UDP rules in each ACL allow DHCP traffic to flow by permitting UDP packets on ports 67 (DHCP server) and 68 (DHCP client). This ensures that devices within the VLANs can obtain IP addresses from a DHCP server, even if other inter-VLAN communication is restricted by the ACLs

```
Home_Router(config)#access-list 100 permit udp any any eq 67
Home_Router(config)#access-list 100 permit udp any any eq 68
Home_Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255
192.168.20.0 0.0.0.255
Home_Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255
192.168.30.0 0.0.0.255
Home_Router(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255
any
Home_Router(config)#access-list 101 permit udp any any eq 67
Home_Router(config)#access-list 101 permit udp any any eq 68
Home_Router(config)#access-list 101 deny ip 192.168.20.0 0.0.0.255
192.168.10.0 0.0.0.255
Home_Router(config)#access-list 101 deny ip 192.168.20.0 0.0.0.255
192.168.30.0 0.0.0.255
Home_Router(config)#access-list 101 permit ip 192.168.20.0 0.0.0.255
any
Home_Router(config)#access-list 102 permit udp any any eq 67
Home_Router(config)#access-list 102 permit udp any any eq 68
Home_Router(config)#access-list 102 deny ip 192.168.30.0 0.0.0.255
192.168.10.0 0.0.0.255
Home_Router(config)#access-list 102 deny ip 192.168.30.0 0.0.0.255
192.168.20.0 0.0.0.255
Home_Router(config)#access-list 102 permit ip 192.168.30.0 0.0.0.255
any
```

```
Home_Router(config)#interface GigabitEthernet0/1.10
Home_Router(config-subif)# ip access-group 100 in
Home_Router(config-subif)#ex
Home_Router(config)#interface GigabitEthernet0/1.20
Home_Router(config-subif)# ip access-group 101 in
Home_Router(config-subif)#ex
Home_Router(config)#interface GigabitEthernet0/1.30
Home_Router(config-subif)# ip access-group 102 in
Home_Router(config-subif)#ex
Home_Router(config)#
Home_Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Home_Router#
```

7. Network Address Translation (NAT) Configuration

Configure NAT for Internet Access

- **Purpose:** Allows devices within the internal network to access the internet using a single public IP address.

```
Home_Router# configure terminal
Home_Router(config)# interface GigabitEthernet0/1
Home_Router(config-if)# ip nat inside
Home_Router(config)# interface GigabitEthernet0/0
```

```
Home_Router(config-if)# ip nat outside
Home_Router(config)# exit
```

Create NAT Access List

- **Purpose:** Define which IP addresses should be translated by NAT.

```
Home_Router(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```

Configure NAT Translation

- **Purpose:** Apply NAT to the access list, enabling overload to support multiple internal devices using a single public IP.

```
Home_Router(config)# ip nat inside source list 1 interface
GigabitEthernet0/0 overload
Home_Router(config)# exit
```

Static Route Configuration:

- **Purpose:** Set a static default route to forward traffic destined for external networks to the ISP's gateway (203.0.113.2).

```
Home_Router(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
Home_Router(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2 10
```

Save Configuration

```
Home_Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

III. Configuring DSL Modem and WAN Emulation Cloud

This shows how to configure a DSL modem and WAN emulation cloud in Cisco Packet Tracer. The setup aims to simulate the connection between a network's edge and an ISP's router.

1. Add one router, one DSL modem, and one WAN emulation cloud (PT Cloud).
2. Connect Devices:
 - Connect the Home DSLModem to Modem4 on Cloud0 using a Phone Line connector in order to simulate realistic DSL internet .
 - Connect the Ethernet port on Home DSLModem to HomeRouter using a Straight-Through Cable to provide the physical connection needed for data transmission between the routers and the modems.
 - Connect the ISP to Cloud0 using the Straight-Through Cable to establish a direct link between the ISP and the WAN emulation cloud, enabling the simulation of an internet connection for the network.
3. Configure ISP:

- **Enter global configuration mode:**

```
Router> enable
Router# configure terminal
Router(config)#hostname ISP_router
```

- **Configure the interface connected to HomeRouter:**

```
ISP_router(config)# interface gigabitEthernet 0/0
ISP_router(config-if)# ip address 203.0.113.2 255.255.255.252
ISP_router(config-if)# no shutdown
ISP_router(config-if)# exit

ISP_router(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

Configure redundancy ISP

```
Router>en
Router#conf t
Router(config)#hostname ISP_2
ISP_2(config)#interface GigabitEthernet0/0
ISP_2(config-if)#description Redundancy_ISP
ISP_2(config-if)#ip address 203.0.114.2 255.255.255.252
ISP_2(config-if)#no shut
ISP_2(config-if)#exit
ISP_2(config)#ip route 0.0.0.0 0.0.0.0 203.0.114.1
ISP_2(config)#exit
ISP_2#copy run start

Destination filename [startup-config]?

Building configuration...
```

- Save the configuration:

```
Router(config)# end
Router# write memory
```

4. Configuring WAN Emulation Cloud

- Select Cloud0 and go to the configuration tab.
- Under the **Connections** section, map Modem 4 to Ethernet 6.

5. Verification

Verify the configurations and connections by pinging between HomeRouter and ISP.

IV. Device Testing and Verification

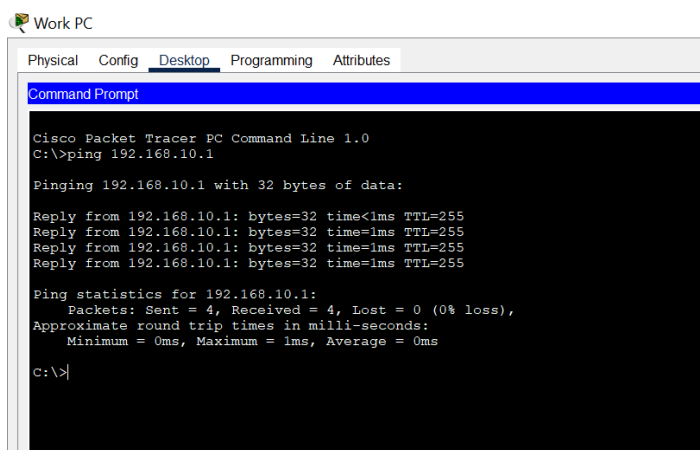
After configuring the network, it's essential to test each device to ensure proper connectivity, VLAN assignment, and functionality. The following steps outline the testing procedure in the network:

- **Default Gateway Testing:**

To ensure proper network communication, the default gateway was tested across various VLANs to verify that it forwards packets correctly and facilitates communication between devices on different networks.

- **Work Devices:**

Action: Pinging the default gateway from a work device in VLAN 10.



```
Work PC
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

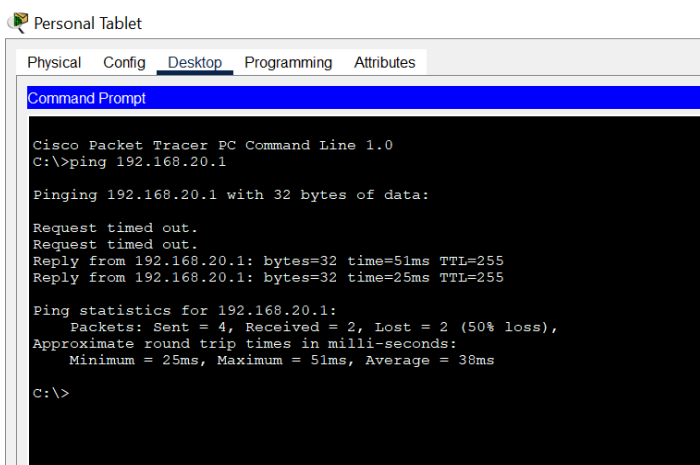
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

- **Personal Devices:**

Action: Pinging the default gateway from a personal device in VLAN 20.



```
Personal Tablet
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.20.1: bytes=32 time=51ms TTL=255
Reply from 192.168.20.1: bytes=32 time=25ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 51ms, Average = 38ms

C:\>
```


Personal Laptop

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=28ms TTL=255
Reply from 192.168.20.1: bytes=32 time=7ms TTL=255
Reply from 192.168.20.1: bytes=32 time=23ms TTL=255
Reply from 192.168.20.1: bytes=32 time=19ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 28ms, Average = 19ms

C:\>|
```

Personal Tablet

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.20.1: bytes=32 time=51ms TTL=255
Reply from 192.168.20.1: bytes=32 time=25ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 51ms, Average = 38ms

C:\>
```

- **Guest Devices:**

Action: Pinging the default gateway from a guest device in VLAN 30.

Guest Laptop

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=32ms TTL=255
Reply from 192.168.30.1: bytes=32 time=34ms TTL=255
Reply from 192.168.30.1: bytes=32 time=23ms TTL=255
Reply from 192.168.30.1: bytes=32 time=31ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 34ms, Average = 30ms

C:\>|
```

Guest Tablet

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=28ms TTL=255
Reply from 192.168.30.1: bytes=32 time=17ms TTL=255
Reply from 192.168.30.1: bytes=32 time=15ms TTL=255
Reply from 192.168.30.1: bytes=32 time=12ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 28ms, Average = 18ms

C:\>|
```

Guest Smartphone

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=34ms TTL=255
Reply from 192.168.30.1: bytes=32 time=13ms TTL=255
Reply from 192.168.30.1: bytes=32 time=21ms TTL=255
Reply from 192.168.30.1: bytes=32 time=22ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 34ms, Average = 22ms

C:\>
```

- **Inter-VLAN Communication:**

Inter- VLAN routing allows devices in different VLANs within the same network to communicate with each other. For security purposes, inter-vlan routing in this project has been blocked via configured access control lists.

Devices in VLAN 10, 20 and 30 are not permitted to communicate with each other

- **Personal Device:**

Personal Laptop

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- Work Device:

Work PC

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.2

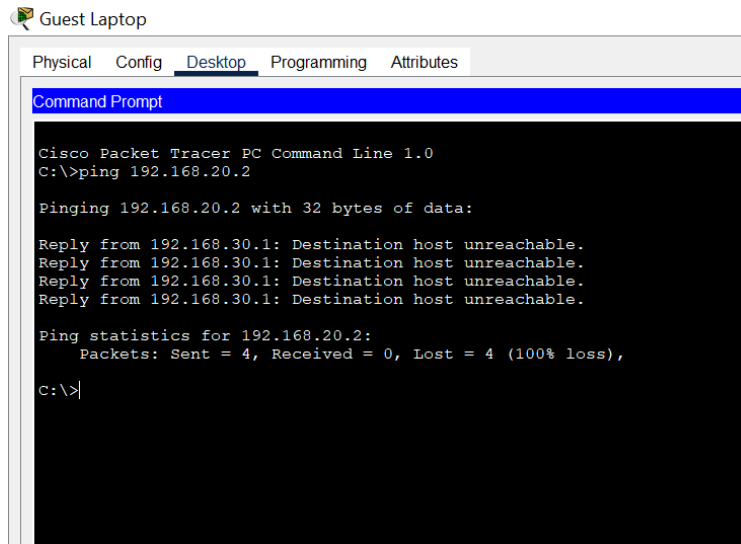
Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- Guest Device:



The screenshot shows a 'Guest Laptop' window with a 'Command Prompt' tab. The text in the command prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

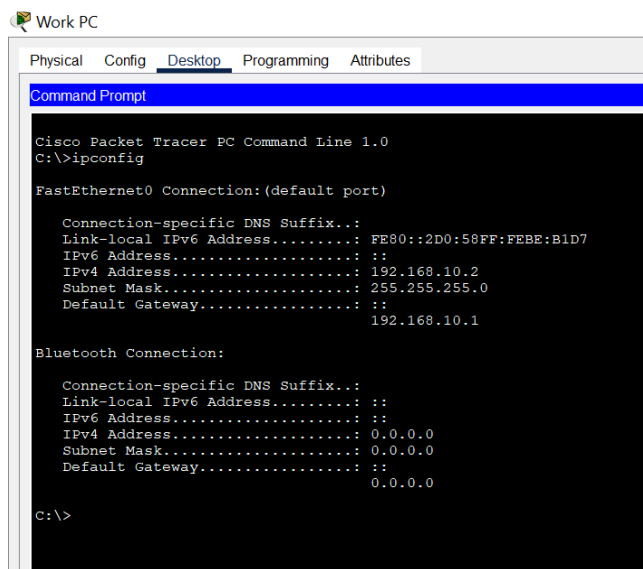
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

• IP Address Verification:

The devices in the network need IP addresses to communicate effectively with each other. To ensure the reliability of network communication, IP address verification was conducted across all VLANs.

- Work Device:



The screenshot shows a 'Work PC' window with a 'Command Prompt' tab. The text in the command prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:58FF:FEBE:B1D7
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

- Personal Devices:

Personal Laptop

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:2FFF:FE74:52E8
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.20.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.20.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
                                0.0.0.0

C:\>
```

Personal Tablet

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20C:85FF:FE5E:10DB
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 192.168.20.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .:
                                192.168.20.1

3G/4G Cell1 Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:8FFF:FE47:DCBB
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
                                0.0.0.0

Bluetooth Connection:
--More--
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
                                0.0.0.0
```

- Guest Devices:

Guest Laptop

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection:(default port)

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::202:16FF:FE01:AB9E
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.30.4
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                192.168.30.1

Bluetooth Connection:

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0

C:\>|
```

Guest Tablet

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection:(default port)

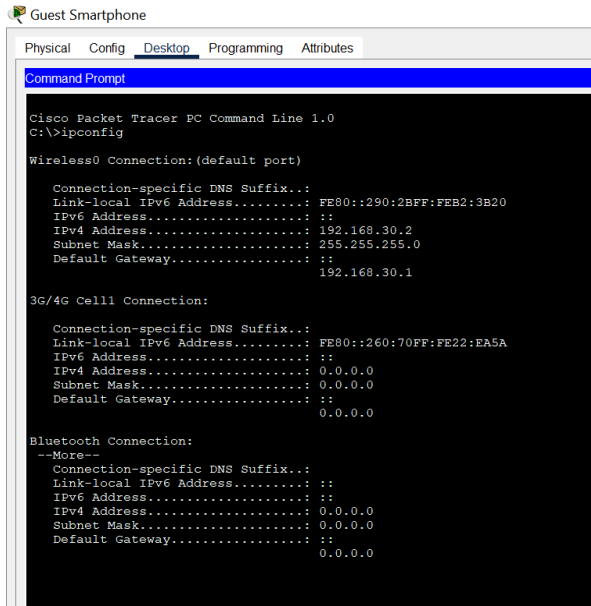
    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::20C:85FF:FE00:E02D
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.30.3
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                192.168.30.1

3G/4G Cell1 Connection:

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::290:21FF:FE4A:DDA0
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0

Bluetooth Connection:
--More--
    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0

|
```



The screenshot shows the 'Guest Smartphone' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the output of the 'ipconfig' command, detailing network configuration for three interfaces: Wireless0, 3G/4G Cell1, and Bluetooth. The Wireless0 interface is configured with a Link-local IPv6 address of FE80::290:2BFF:FE2:3B20, an IPv4 address of 192.168.30.2, and a default gateway of 192.168.30.1. The 3G/4G Cell1 and Bluetooth interfaces are both configured with 0.0.0.0 for all IP-related fields.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection:(default port)

Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: FE80::290:2BFF:FE2:3B20
IPv6 Address.....: ::
IPv4 Address.....: 192.168.30.2
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.30.1

3G/4G Cell1 Connection:

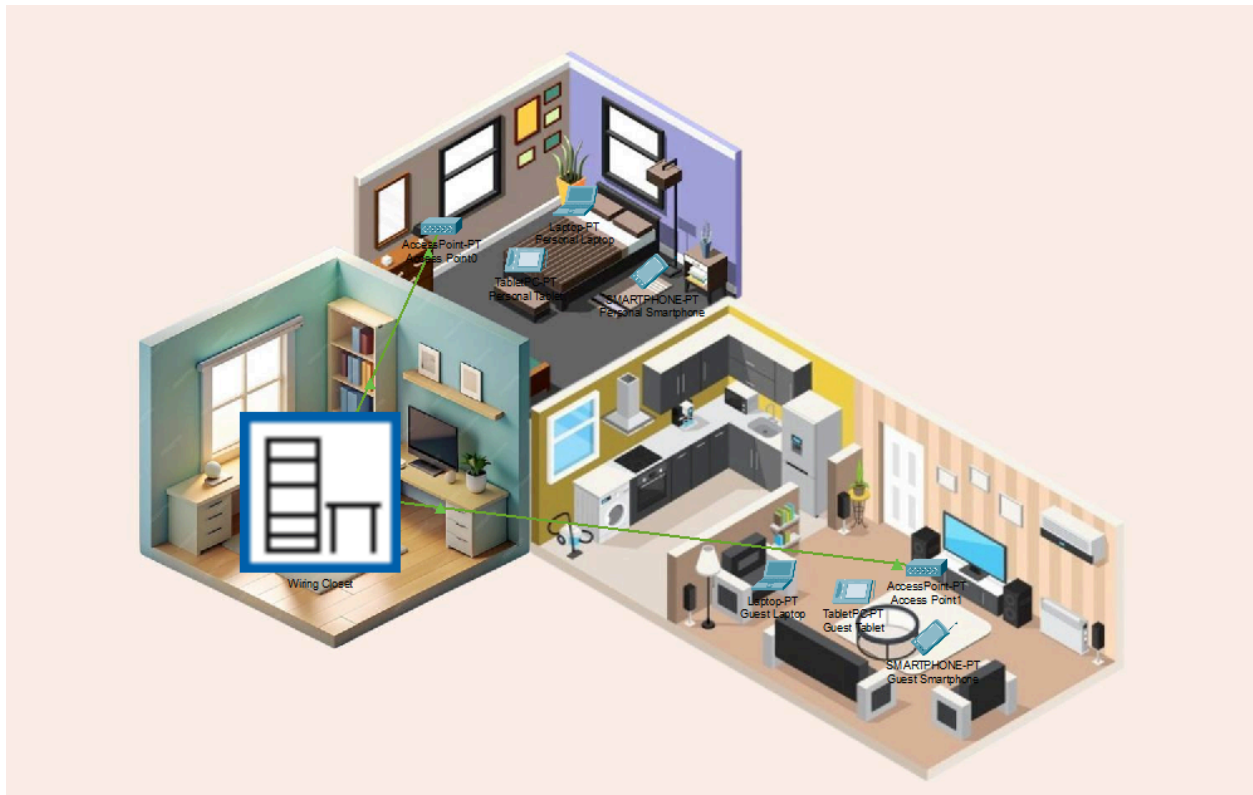
Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: FE80::260:70FF:FE22:EA5A
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0

Bluetooth Connection:
--More--
Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0
```

This document has provided a comprehensive guide to setting up a home office network using Cisco Packet Tracer. The setup involved configuring VLANs, inter-VLAN routing, DHCP services, wireless security, access control lists (ACLs), and Network Address Translation (NAT). Additionally, the integration of a DSL modem and WAN emulation cloud was demonstrated to simulate real-world ISP connectivity.

Low-Level Documentation for Home Office Network Configuration

This document outlines the physical setup of a home office network using Cisco Packet Tracer.



Assumptions and Constraints

- **Assumptions:**

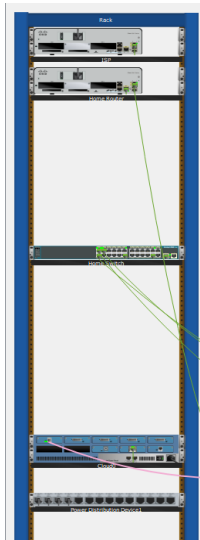
- All required network devices are available in Cisco Packet Tracer.
- The network will be implemented in a home office environment, simulating a small-scale network with multiple VLANs.
- The wiring closet in the office of the home serves as the home office of the client.

- **Constraints:**

- The power supply must support all network devices without interruption.
- The network simulation assumes a stable ISP connection via the WAN emulation cloud (Cloud0) for accurate testing and performance evaluation.

Diagrams of Rack, Table, and Slot Layouts

Rack Layout Overview



Top Shelf:

- **ISP Router (Cisco 1941):**
 - This router simulates the ISP connection, acting as the gateway to the internet.
- **Home Router (Cisco 1941):**

- Positioned on the top shelf to serve as the central gateway for the network.
- Manages inter-VLAN routing and connections to the ISP.
- This location minimizes cable clutter, offering easy access for connections to other networking devices.

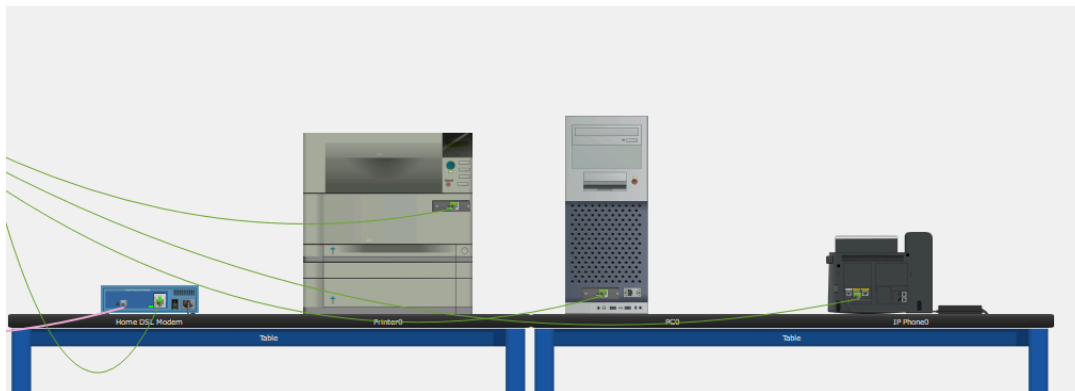
Middle Shelf:

- **Home Switch (Cisco 2960):**
 - Centrally located to efficiently manage connections to devices across different VLANs.

Bottom Shelf:

- **Cloud Device (Cloud0):**
 - Positioned on the bottom shelf to maintain organization and reduce heat buildup.
- **Power Distribution Device:**
 - Placed on the bottom shelf due to its weight, lowering the center of gravity and stabilizing the rack.
 - Allows easy access to power outlets, facilitating efficient cable management.

Table Layout Overview



- **Printer:** Positioned to the left side of the table for easy access when printing documents.
- **PC:** Centrally located on the table, serving as the primary workstation.
- **DSL Modem:** Placed close to the power outlet, and connected to the router for internet access.
- **Printer:** Positioned to the right side of the table, within reach for quick communication.

Slot Layouts



ISP and Home router

GigabitEthernet 0/0 (ISP Router):

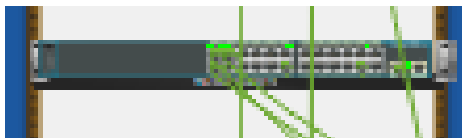
- **Connection:** Cloud0.
- **Purpose:** Provides simulated internet connectivity by linking the ISP router to the WAN emulation cloud.

GigabitEthernet 0/0 (Home Router):

- **Connection:** ISP via Cloud0.
- **Purpose:** Provides internet connectivity by linking the home router to the ISP router through the cloud simulation.

GigabitEthernet 0/1 (Home Router):

- **Connection:** Home Switch.
- **Purpose:** Handles internal network traffic by connecting the router to the home switch.



Switch



Access Points

FastEthernet 0/1-10 (Switch):

- **Assigned to:** VLAN 10 (Work Devices).
- **Purpose:** Connects work-related devices to the network.

FastEthernet 0/11-20 (Switch):

- **Assigned to:** VLAN 20 (Personal Devices).
- **Purpose:** Connects personal devices to the network.
- **Connected Device:** Access Point (for Personal Devices) FastEthernet 0/11 (Switch).

FastEthernet 0/21-24 (Switch):

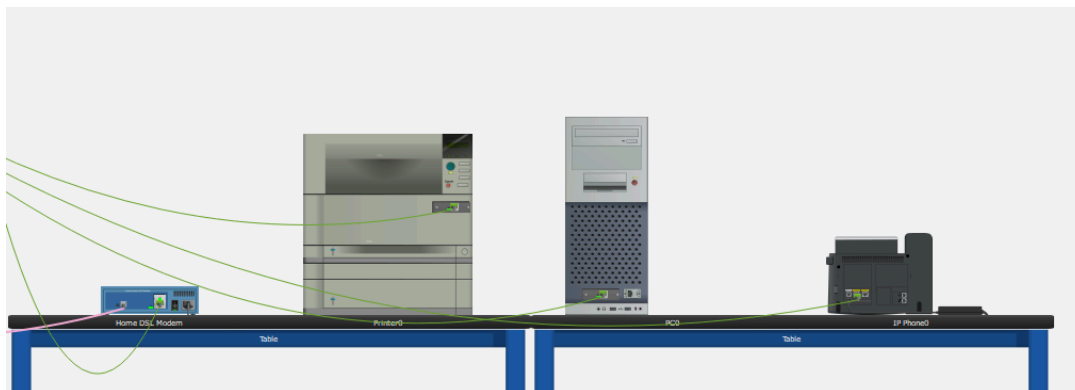
- **Assigned to:** VLAN 30 (Guest Devices).
- **Purpose:** Connects guest devices to the network.
- **Connected Device:** Access Point (for Guest Devices) FastEthernet 0/22 (Switch).

GigabitEthernet 0/1 (Switch):

- **Connection:** Trunk to Home Router.
- **Purpose:** Carries traffic for multiple VLANs between the switch and the router.

End Device Layouts

Wired Connections

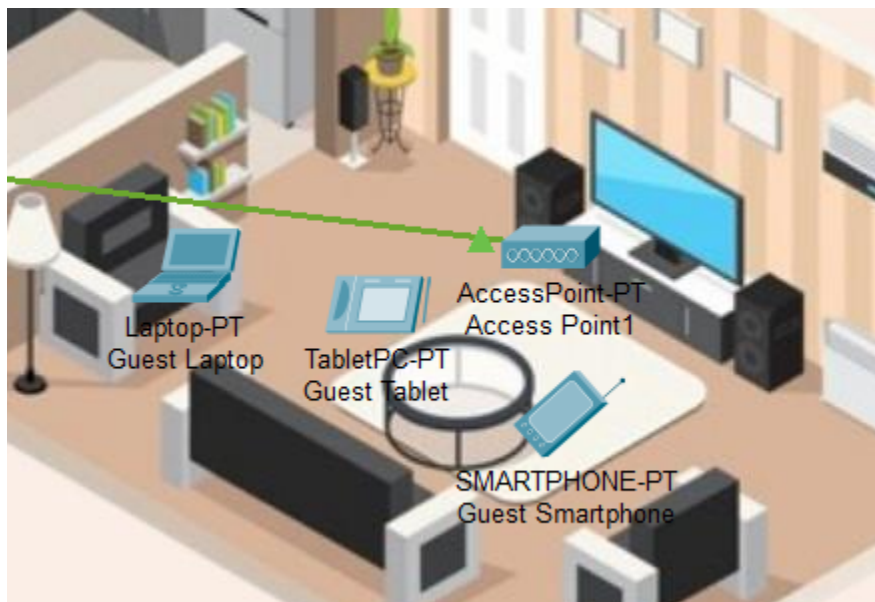


1. **PC (Work Device)**
 - **VLAN:** 10 (Work Devices)
 - **Connection:** FastEthernet 0/1 (Switch)
 - **Location:** On the table
2. **Printer (Work Device)**
 - **VLAN:** 10 (Work Devices)

- **Connection:** FastEthernet 0/2 (Switch)
 - **Location:** On the table
3. **Phone (Work Device)**
- **VLAN:** 10 (Work Devices)
 - **Connection:** FastEthernet 0/3 (Switch)
 - **Location:** On the table
4. **DSL Modem**
- **VLAN:** Not assigned (Connected directly to ISP)
 - **Connection:** GigabitEthernet 0/0 (Home Router)
 - **Location:** On the table

Wireless Connections

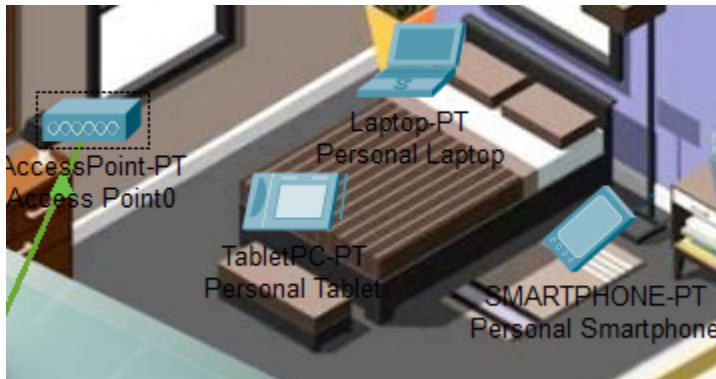
1. Access Point 1 (Personal Devices)



- **VLAN:** 20 (Personal Devices)
- **Connection:** FastEthernet 0/11 (Switch)
- **Purpose:** Provides wireless connectivity to personal devices.
- **Connected Devices:**
 - **Tablet**
 - **Smartphone**
 - **Laptop**

- **Location:** In the bedroom

2. Access Point 2 (Guest Devices)



- **VLAN:** 30 (Guest Devices)
- **Connection:** FastEthernet 0/22 (Switch)
- **Purpose:** Provides wireless connectivity to guest devices.
- **Connected Devices:**
 - **Guest Laptop**
 - **Guest Smartphone**
 - **Guest Tablet**
- **Location:** In the living room

Additional Notes:

- **Device Placement:**
 - The table hosts the DSL modem, PC, printer, and phone, which are all connected via Ethernet cables. The table's proximity to the switch and router is essential to minimize cable lengths and reduce signal interference.
- **Wireless Device Distribution:**
 - Devices connected to the Access Points are strategically positioned around the home to ensure optimal wireless coverage. Access Point 1, located in the bedroom, caters to personal devices. This placement provides the best coverage for the client, who frequently uses personal devices in this area. Access Point 2, placed in the living room, is reserved for guest devices. This location ensures that

guests have fast and reliable internet connectivity, keeping traffic segmented and secure.

This low-level documentation details the carefully planned setup of a home office network using Cisco Packet Tracer. The configuration is designed to optimize both wired and wireless connectivity within a small-scale environment, ensuring efficient operation and ease of use.

Key components, including the routers, switch, access points, and end devices, have been strategically placed to minimize cable clutter and enhance network performance. The use of VLANs effectively segments traffic between work, personal, and guest devices, providing both security and organization within the network.

The placement of Access Points in critical areas of the home—such as the bedroom for personal devices and the living room for guest devices—ensures optimal wireless coverage and a seamless user experience.

The home office network is equipped to handle daily tasks efficiently, providing a reliable, secure, and scalable networking solution tailored to the client's needs.